

Windows 11 (Home & Pro): Telemetry, Tracking, and Spyware-Equivalent Functionality — A Forensic Technical Analysis

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Part 1: Title & Executive Summary

Windows 11 (Home & Pro): Telemetry, Tracking, and Spyware-Equivalent Functionality — A Forensic Technical Analysis

Version: 1.0 **Date:** November 2025

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Purpose: To document and assess Windows 11 features and mechanisms which, under technical definitions of spyware behaviour, could be construed as user-monitoring or covert data collection.

Executive Summary

This report provides a forensic-level analysis of Windows 11 Home and Pro editions, identifying all known telemetry, tracking, profiling, and data-transmission mechanisms which resemble spyware behaviour under the operational definition of the term.

While Microsoft asserts that telemetry serves diagnostic and product-improvement purposes, the volume, persistence, and opacity of these data flows create privacy concerns analogous to enterprise-grade surveillance software.

Data is collected across multiple system layers — core OS services, cloud integration, Microsoft applications, and bundled third-party components.

Full deactivation of these channels is not possible in Home and Pro editions; limited mitigation can be achieved through Group Policy, Registry, and firewall configuration.

Part 2: Definition & Methodology

Spyware Definition Used

Software qualifies as spyware-equivalent if it:

1. Collects or transmits user data without explicit, informed consent.
2. Operates persistently or re-enables after user attempts to disable it.
3. Lacks transparency in scope or destination of collected data.
4. Correlates system data with user identifiers for profiling or marketing.

Methodology

- Reverse-engineering of Windows 11 22H2–24H2 processes and services.
- Analysis of network telemetry endpoints.
- Policy and registry documentation (Microsoft ADMX templates).
- Comparison with enterprise-grade diagnostic systems.

Part 3: Core Telemetry Infrastructure

Component	Executable/Service	Data Collected	Destination
Connected User Experiences and Telemetry Service (DiagTrack)	DiagTrack.dll / svchost.exe	App usage, crash dumps, reliability metrics, system state	vortex-win.data.microsoft.com, settings-win.data.microsoft.com
dmwappushservice	dmwappushsvc.dll	Push notifications and diagnostics	Microsoft Push Services
Customer Experience Improvement Program (CEIP)	Task Scheduler tasks	App usage statistics	Microsoft CEIP servers
Windows Error Reporting (WER)	WerFault.exe	Memory dumps, user context of crashes	watson.telemetry.microsoft.com
Compatibility	compattelrunner.exe	App	Microsoft compatibility endpoints

Component	Executable/Service	Data Collected	Destination
Telemetry		inventory, driver data	

Part 4: System Subsystems & Components

4.1 Windows Search and Start Menu

- Sends typed queries to Bing (`api.bing.microsoft.com`) for cloud search.
- Collects keystroke and interaction data.
- Mitigation: disable *Search online and include web results* (partial).

4.2 Microsoft Edge Browser

- Sends telemetry to `self.events.data.microsoft.com`.
- Syncs browsing history, autofill, and telemetry unless disabled.
- Advertising ID and Microsoft account identifiers used for profiling.

4.3 Microsoft Defender & SmartScreen

- Hashes of executed files sent to `wdcp.microsoft.com` and `smartscreen.microsoft.com`.
- Legitimate for threat detection but resembles behavioural monitoring.

4.4 OneDrive Integration

- Background sync service (`onedrivesetup.exe`, `onedrive.exe`).
- Collects file metadata and activity for cloud sync.
- Automatically re-enabled after updates.

4.5 Widgets & News Feed

- Fetches personalised content from MSN/Bing endpoints.
- Uses Microsoft Advertising ID and browsing interests.

Part 5: Third-Party & Bundled Integrations

Integration	Function	Data Collected	Notes
Xbox Game Bar & Xbox Services	Gameplay capture, performance data	Game telemetry and activity	Can be uninstalled or blocked via firewall
OEM Telemetry Agents	Dell, HP, Lenovo, etc.	Device health and usage metrics	Often use separate endpoints (e.g. hpccp.hp.com)
Microsoft Store Apps (Mail, Calendar, etc.)	App telemetry & user activity	Shared with Microsoft's App Center analytics	Remove or restrict permissions
Bing Copilot / AI Assistant	Prompt and usage logs	Interaction data, possibly local context	Feature cannot be fully disabled in Home

Part 6: Network Endpoint Summary

Known telemetry domains contacted by default installations (Windows 11 24H2, Home edition):

```
vortex-win.data.microsoft.com
settings-win.data.microsoft.com
watson.telemetry.microsoft.com
self.events.data.microsoft.com
telecommand.telemetry.microsoft.com
oca.telemetry.microsoft.com
tsfe.trafficshaping.dsp.mp.microsoft.com
browser.events.data.msn.com
```

Blocking via firewall or hosts file may impair updates or Store functionality.

Part 7: Mitigation & Control

Mitigation Layer	Method	Effectiveness
Local Policy / Registry	Set AllowTelemetry = 0 (HKLM\Software\Policies\Microsoft\Windows\DataCollection)	Partial; Home ignores policy
Services Management	Disable DiagTrack, dmwappushservice	Stops core telemetry temporarily
Firewall / DNS Blocking	Block listed endpoints	Effective but may affect Windows Update
Local Account Use	Avoid Microsoft Account sign-in	Reduces cloud linkage
Third-Party Tools	O&O ShutUp10++, WPD, PrivateZilla	Centralize disabling of privacy-related tasks

Part 8: Discussion

While Microsoft categorizes these functions as diagnostic or security-related, forensic review reveals behaviour consistent with spyware definitions: persistent data transmission, correlation with user identity, and difficulty of complete deactivation.

Windows 11 Home and Pro provide limited control; only Enterprise/Education editions allow near-complete suppression of telemetry.

Part 9: References

- Microsoft Privacy Whitepaper (2024): *Diagnostic, feedback, and telemetry in Windows*
 - Dataflow analysis by German BSI (Bundesamt für Sicherheit in der Informationstechnik), 2023
 - Ars Technica: “Windows 10/11 Telemetry Deep Dive”
 - O&O Software: ShutUp10++ Telemetry Map (2024)
 - Independent captures from Wireshark network sessions (2022-2025)
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Part 10: Conclusion

Windows 11 integrates extensive telemetry at the OS core, combining diagnostic, behavioural, and marketing data streams.

Although not classified by Microsoft as spyware, under the operational definition used in this paper, multiple subsystems exhibit **spyware-equivalent characteristics** due to:

1. Lack of complete user opt-out,
2. Persistent data retransmission even after disabling attempts,
3. Data correlation with identifiable accounts.

Mitigation is possible but partial; true privacy requires isolation (e.g., firewall blocking or network segmentation) or migration to Enterprise builds with telemetry = 0.

End of report text